# Generate BBH Initial Data Using Physics Informed DeepONet (PIDONet)

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# Issues with Physics Informed FNO

FNO requires Fourier transforming an entire grid of data

- TwoPunctures generate data in a spherical domain
- Some datapoints are removed to be used for validation

Trained PINO can only predict correction field for a rectangular domain

- Cannot validate against individual points or test against a spherical domain
- Cannot generate high-resolution datasets due to memory constraints
  - TwoPunctures can generate datasets in 1D or 2D domains

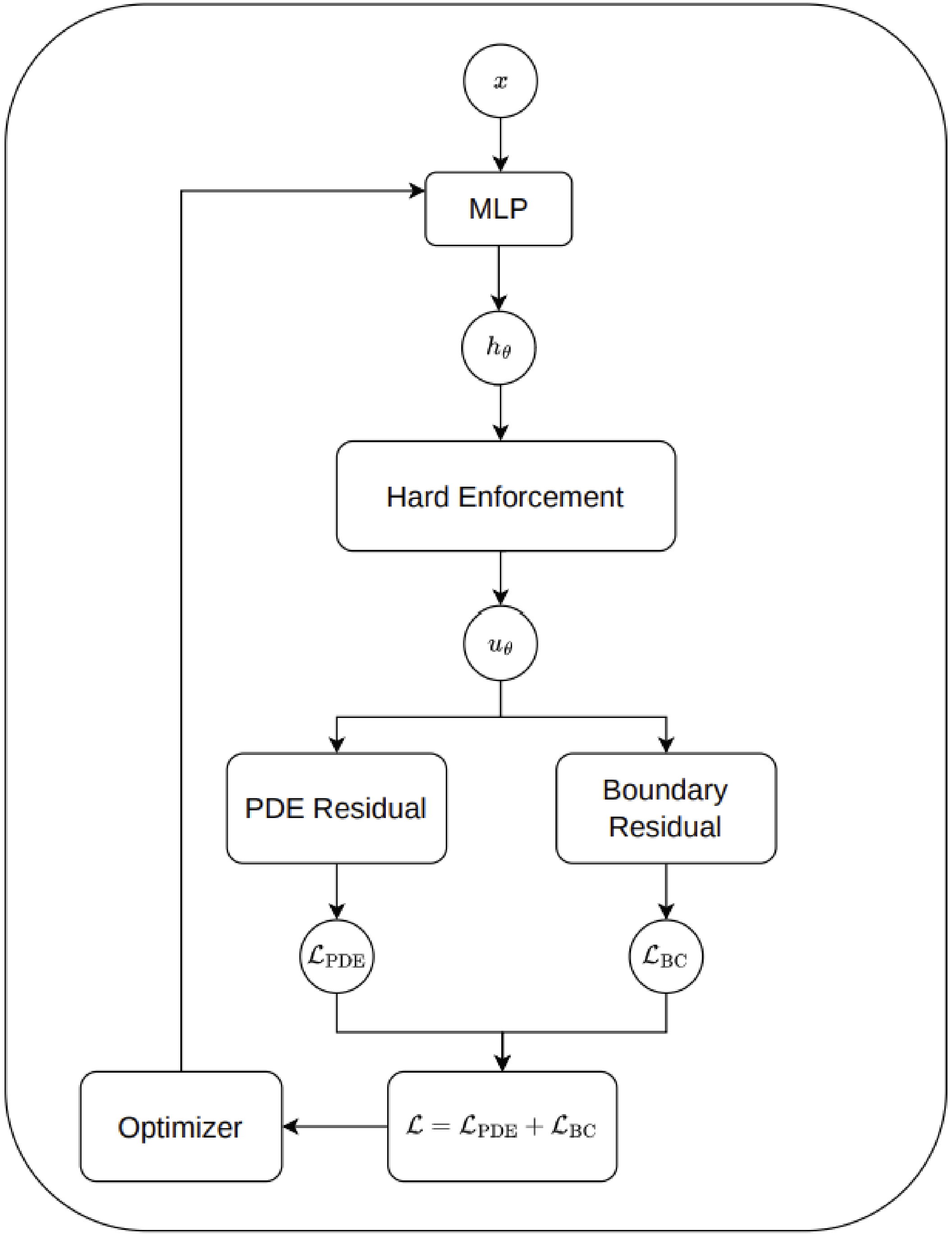
# Theory

Einstein's Field Equations:

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}$$

Hamiltonian Constraint Equation:

$$\Delta \mathrm{u} + \frac{1}{8}\left(1 + \sum_{n} \frac{m_n}{2r_n} + \mathrm{u}\right)^{-7} \bar{K}_{ij}\bar{K}^{ij} = 0$$



$x$
MLP
$h_\theta$
Hard Enforcement
$u_\theta$
PDE Residual
Boundary Residual
$\mathcal{L}_{\mathrm{PDE}}$
$\mathcal{L}_{\mathrm{BC}}$
Optimizer
$\mathcal{L} = \mathcal{L}_{\mathrm{PDE}} + \mathcal{L}_{\mathrm{BC}}$

# Loss function

$$\mathcal{L} = w_2 L_2 + w_\infty \text{soft-}L_\infty + w_{rob} \mathcal{L}_{BC}$$

$L_2 \sim 10^{-8} - 10^{-11}$

soft-$L_\infty \sim 10^{-5}$

$\mathcal{L}_{BC} \sim 10^{-12}$

# Exponential Moving Average (EMA) method

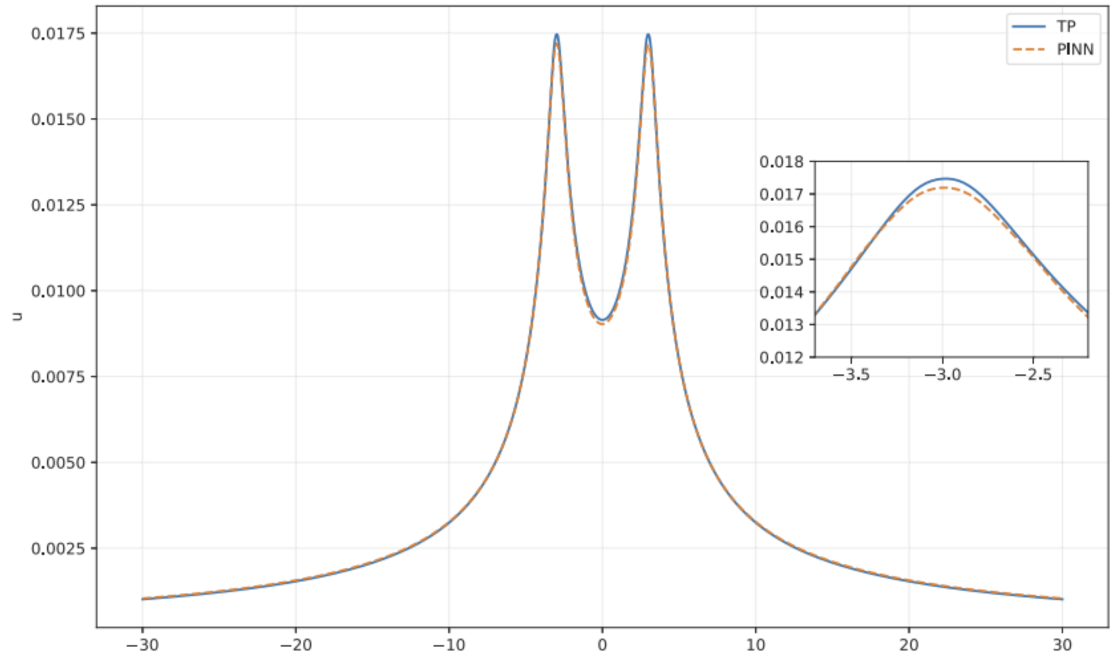
$$\bar{L}_k(t_n) = \alpha \bar{L}_k(t_{n-1}) + (1 - \alpha) L_k(t_n)$$

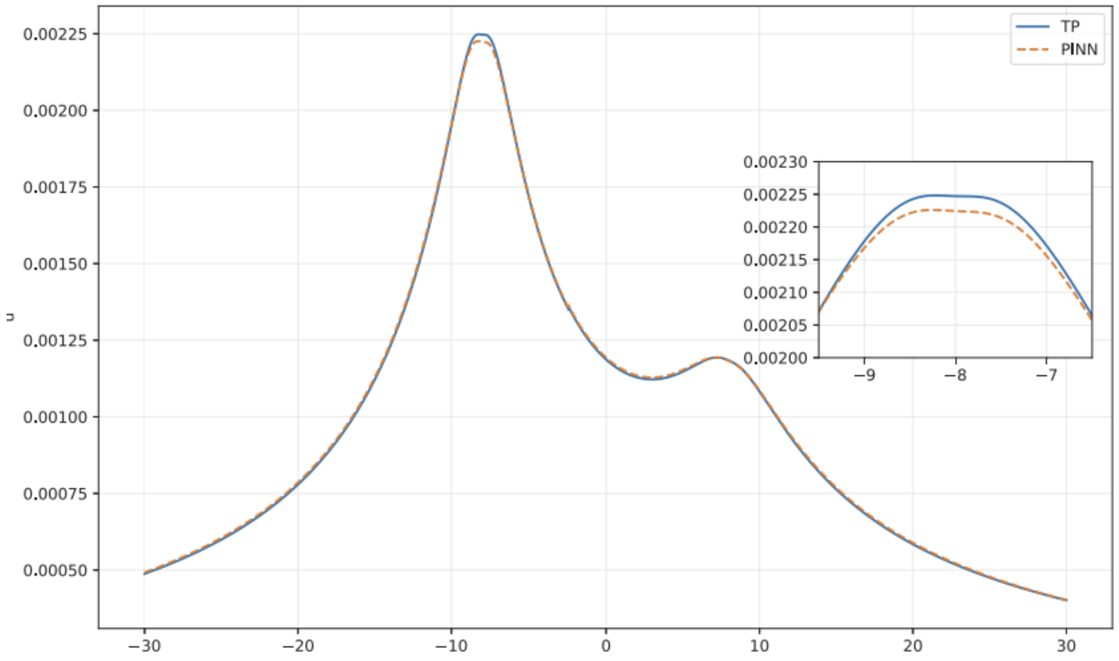
$$L_k = \{L_2, \text{soft-}L_\infty, \mathcal{L}_{BC}\}$$

$\alpha \in [0,1)$, chosen to be 0.99

$$\tilde{L}(t_n) = \frac{L_k(t_n)}{\bar{L}_k(t_n)}$$

# Paper Results

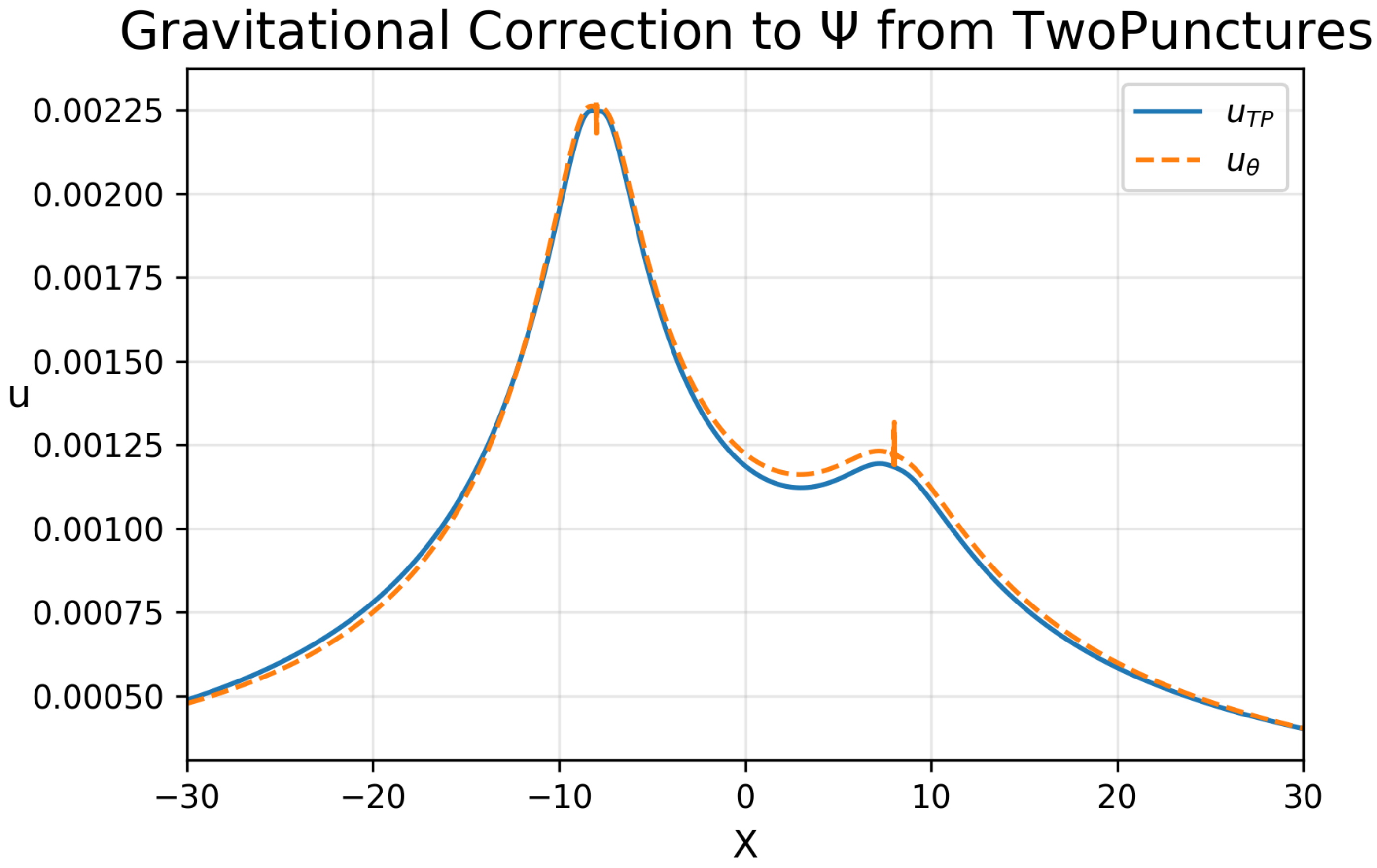



$L2RE = 0.0172 \pm 0.0040$

$L2RE = 0.0077 \pm 0.0007$

# First attempt



# Ansatz

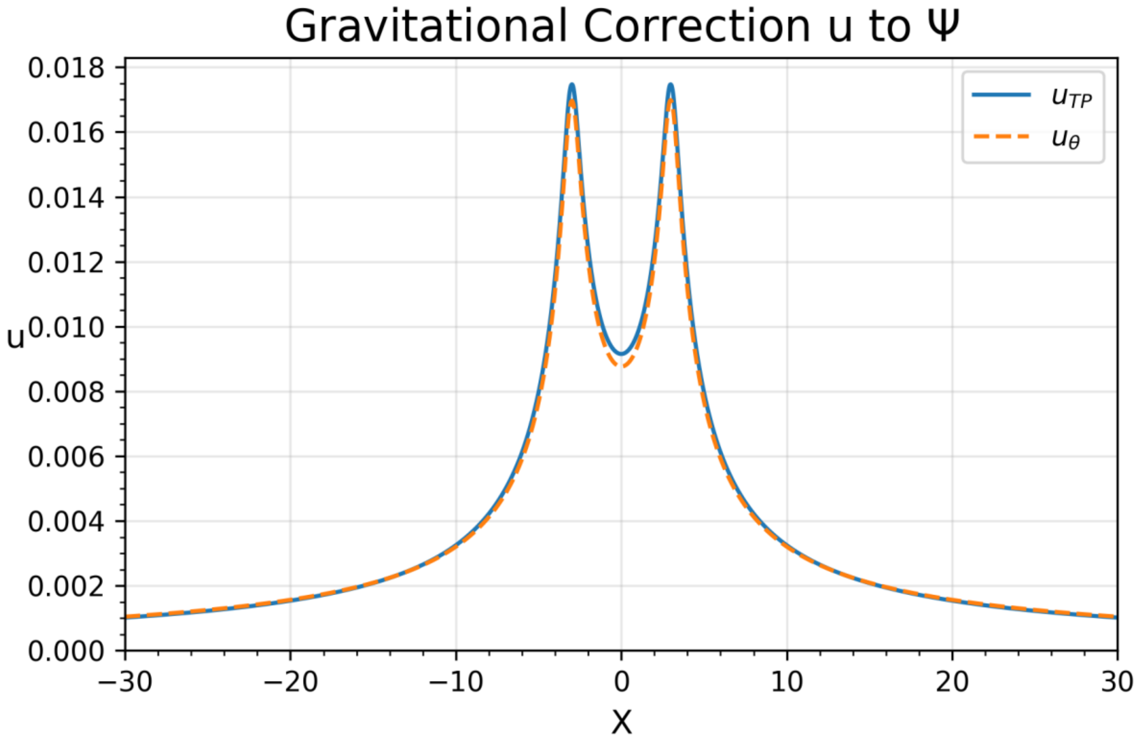
$$u_\theta(x) = \kappa u_g(x)\left[1 + cW(x)\tanh\left(h_\theta(x)\right)\right]$$

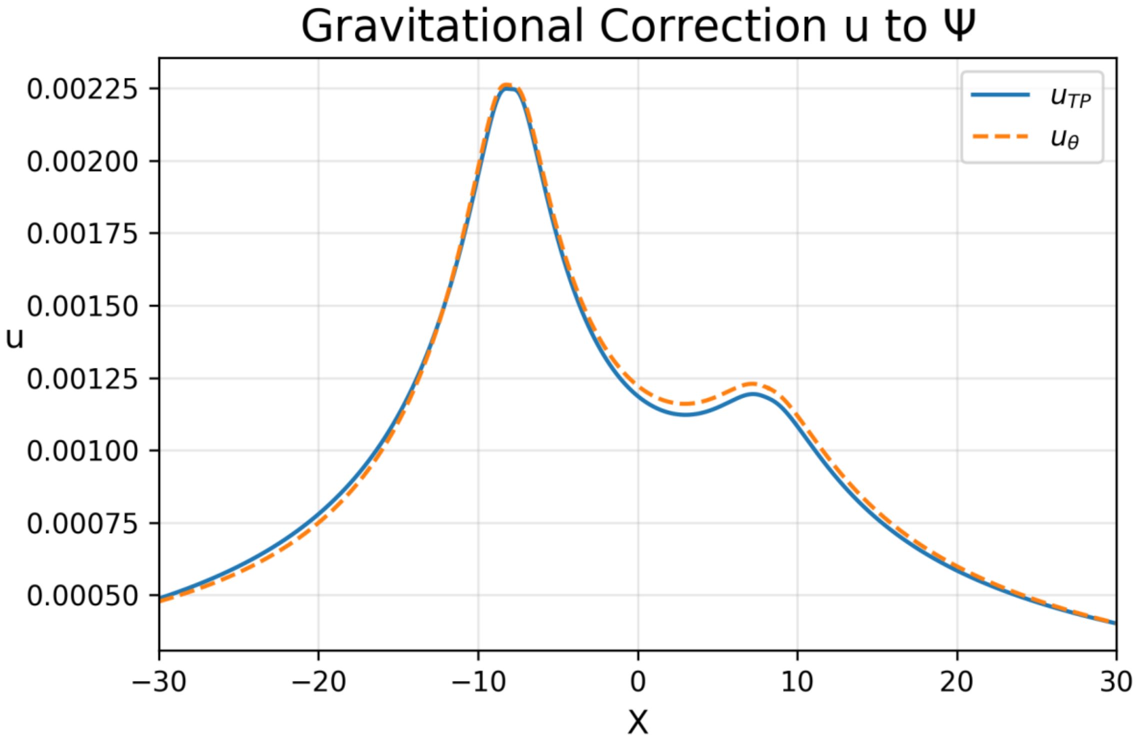
$$u_2 = \frac{15l + 132l^2 + 53l^3 + 96l^4 + 82l^5 + \frac{84l^5}{\mathcal{R}} + \frac{85\ln l}{\mathcal{R}^2}}{80\mathcal{R}}$$

$$\mathcal{R} = \frac{2r_n}{m}, l = \frac{1}{1+\mathcal{R}}$$

Taylor expand to $\mathcal{O}(10^{-14})$ for $r_n < 0.1$ to sidestep zero-error

# Second attempt

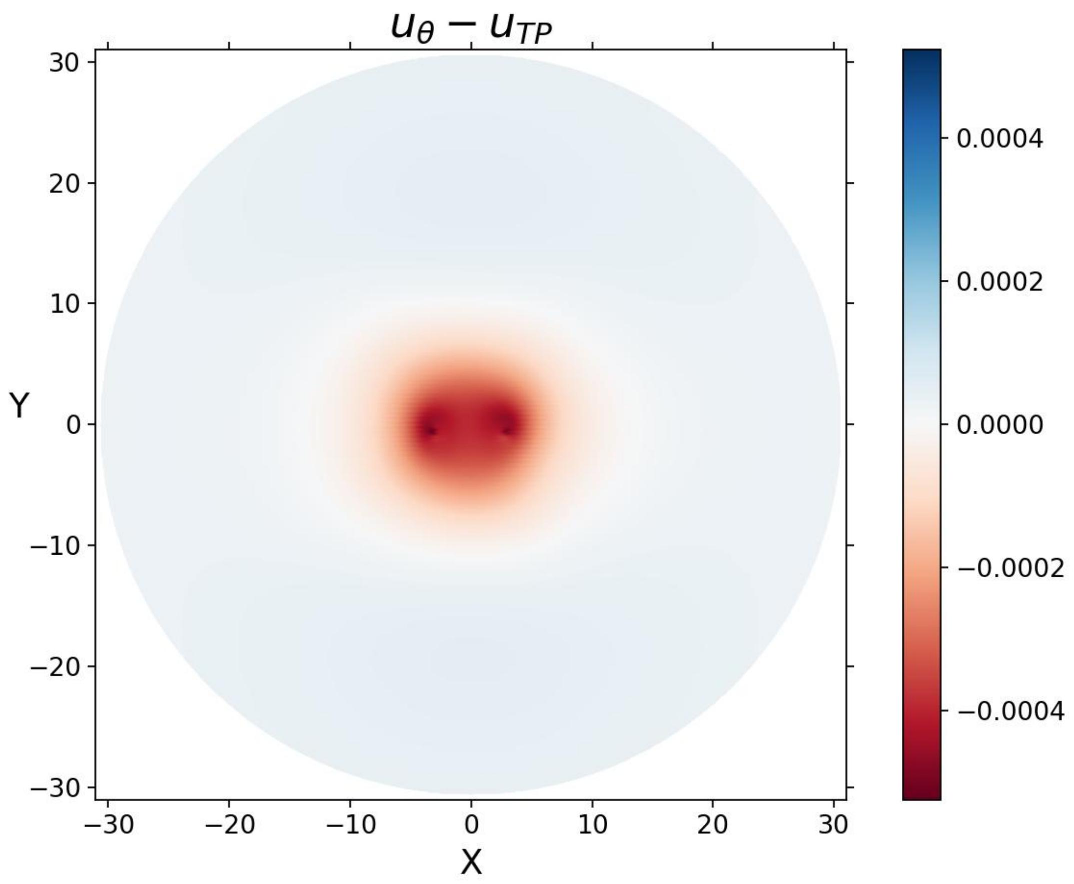


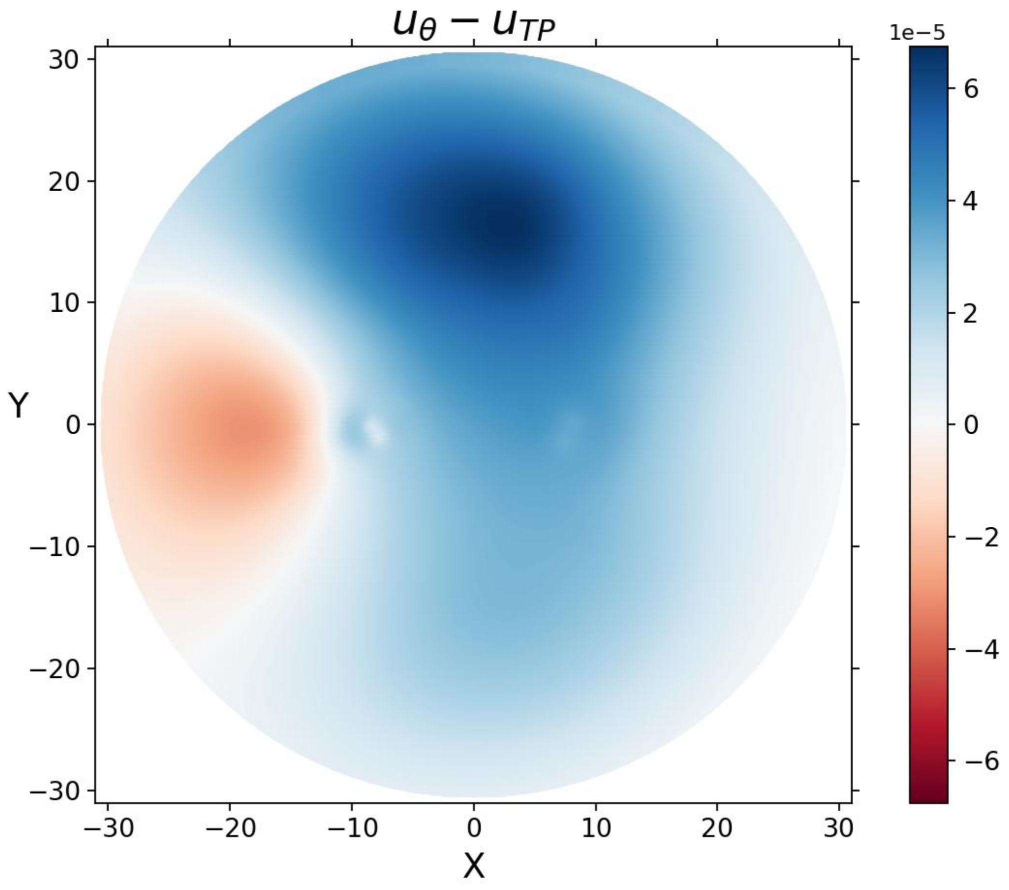

$L2RE = 0.028$

$L2RE = 0.048$

# In XY-plane

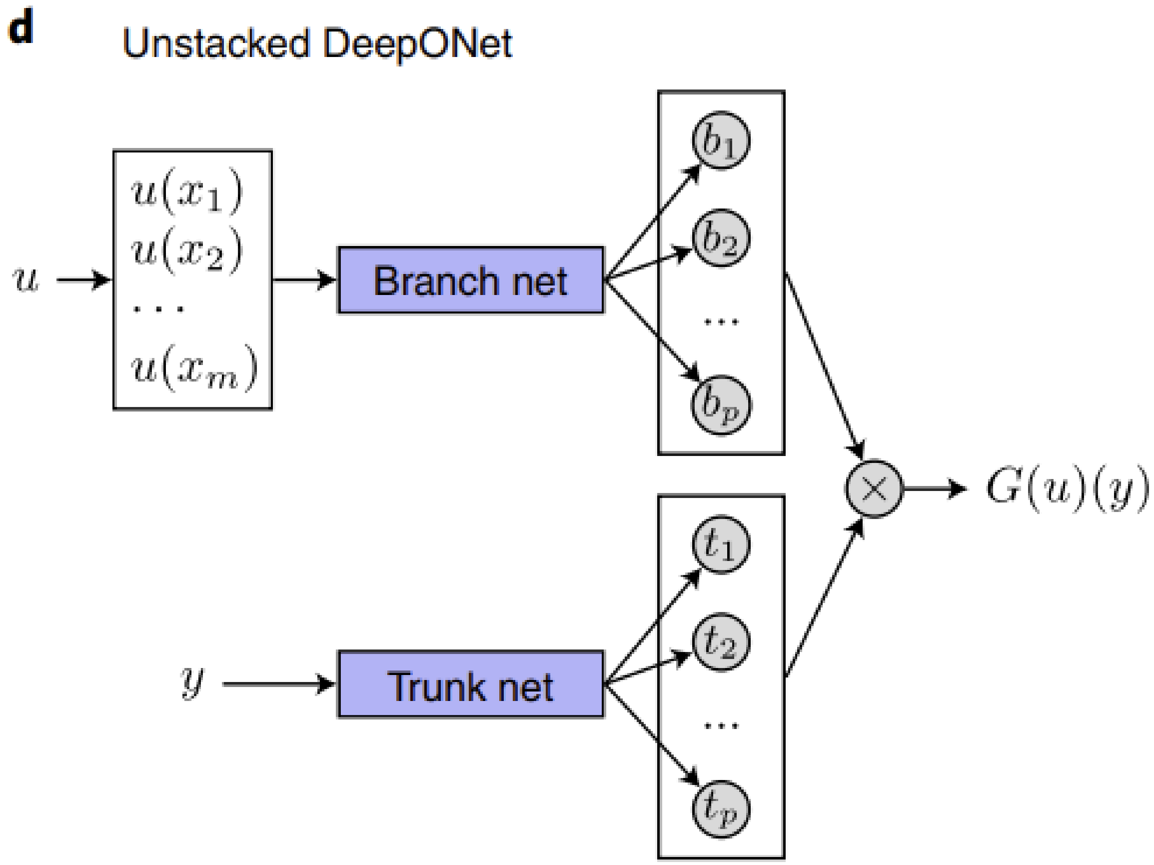


$u_\theta - u_{TP}$
Y
X
30
20
10
0
−10
−20
−30
0.0004
0.0002
0.0000
−0.0002
−0.0004

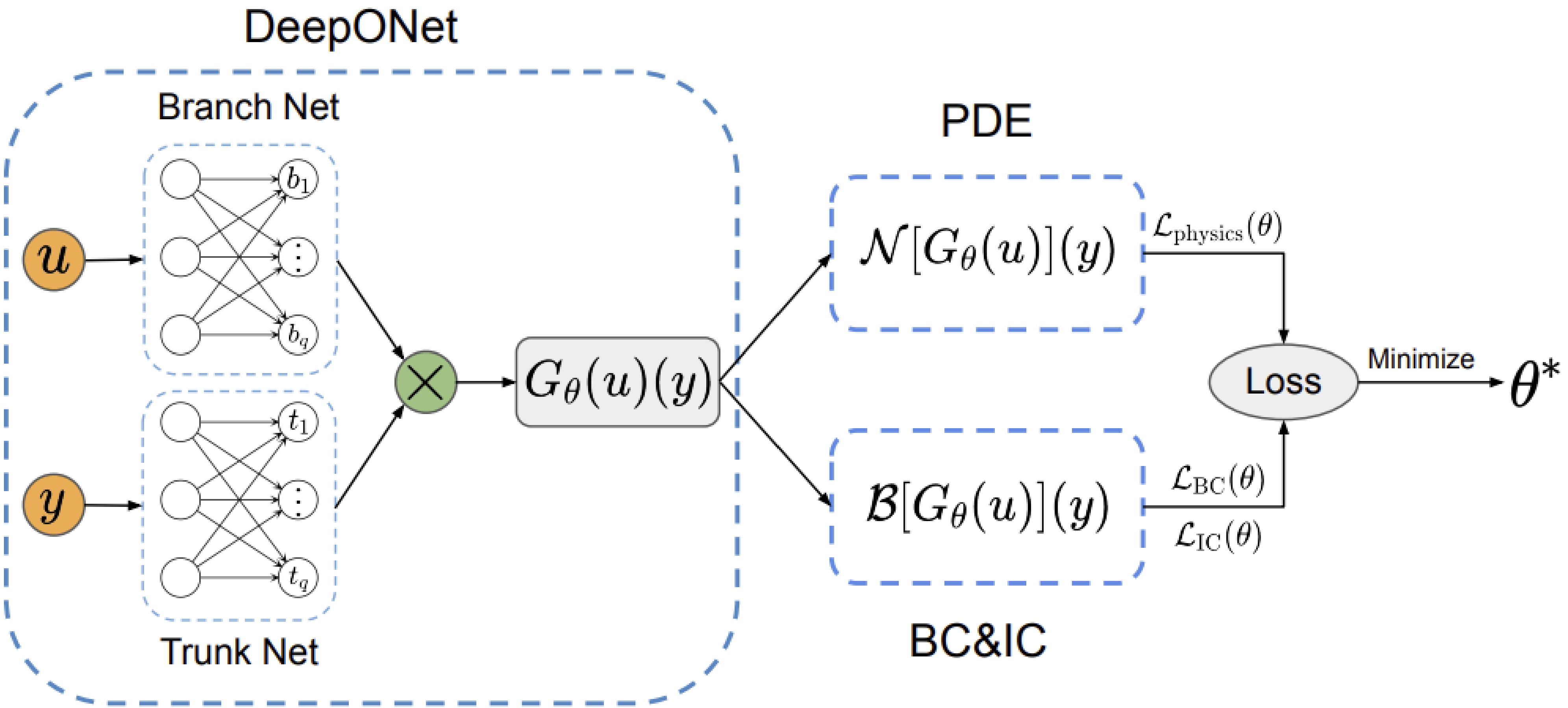
$u_\theta - u_{TP}$
Y
X
30
20
10
0
−10
−20
−30
1e−5
6
4
2
0
−2
−4
−6

# DeepONet Recap




$$G(u)(y) = \sum_{k=1}^{p} b_k \, t_k + b_0$$

# Physics Informed DeepONet (PIDONet)



# Project Plan

Train PIDONet to generate BBH initial data for $m_+ = [0.15, 0.25, 0.35, 0.45]$

- ~ 37 million datapoints from TwoPunctures for $m_+ = [0.1, 0.2, 0.3, 0.4, 0.5]$
  - ~30 million datapoints used for training
  - ~ 7 million datapoints used for validation
- 2 million interior datapoints to calculate PDE loss
  - 1 million datapoints restricted to $R < 10$
  - 1 million datapoints for $10 < R < 30$
- 200000 boundary datapoints to enforce boundary conditions
- Collocation points sampled from $m_+ \in [0.1, 0.5]$

PIDONet to predict $h_\theta$

Ansatz to predict $u_\theta$

# Operator Training Architecture

Branch net input:

- $b_k = [m_+, m_-, x_+, y_+, z_+, x_-, y_-, z_-, P_{x+}, P_{y+}, P_{z+}, P_{x-}, P_{y-}, P_{z-}]$
- $m_+ + m_- = 1$
- $X_+ = [3,0,0]$, $X_- = [-3,0,0]$
- $P_{y+} = 0.8 \times m_+ \times m_-$, $P_{y+} = -P_{y-}$, $P_{x+} = P_{z+} = P_{x-} = P_{z-} = 0$

Trunk net input:

- $t_k = [x, y, z]$

Architecture:

- 3 hidden layers
- 64 neurons each
- SiLU activation function
- 50 outputs

# Loss function

1. L2 error on datapoints
2. L2 error on interior points
3. soft-$L_\infty$ error on interior points
4. L2 error on boundary points

$$\mathcal{L} = L_{data} + L_{int} + 0.5 \times L_\infty + L_{RB}$$

Losses rescaled using EMA method

Adam optimizer ($lr = 10^{-4}$)

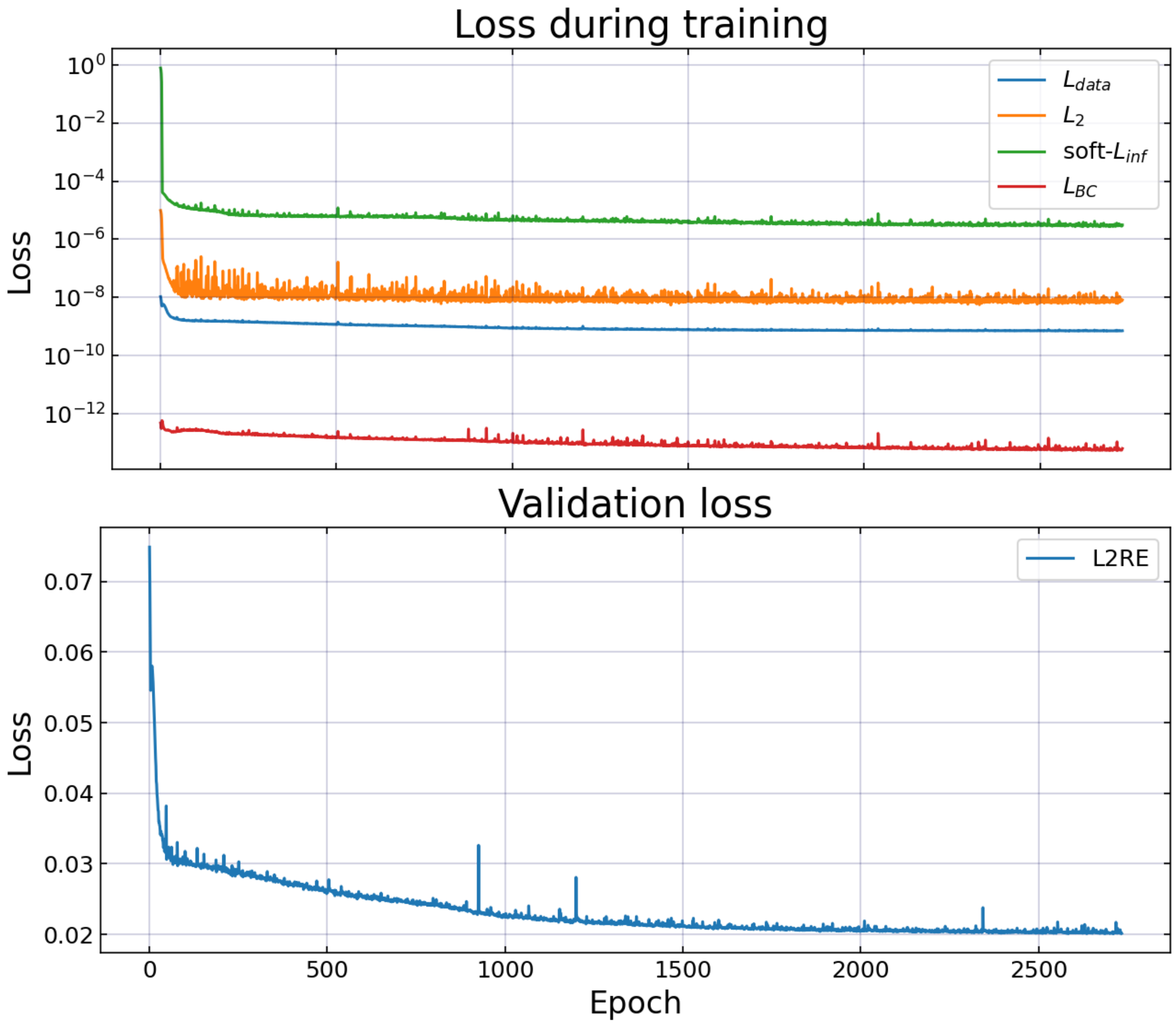
L2RE on datapoints for validation

# Results

Trained for 2734 epochs (8 hours 20 minutes on A100 GPU)

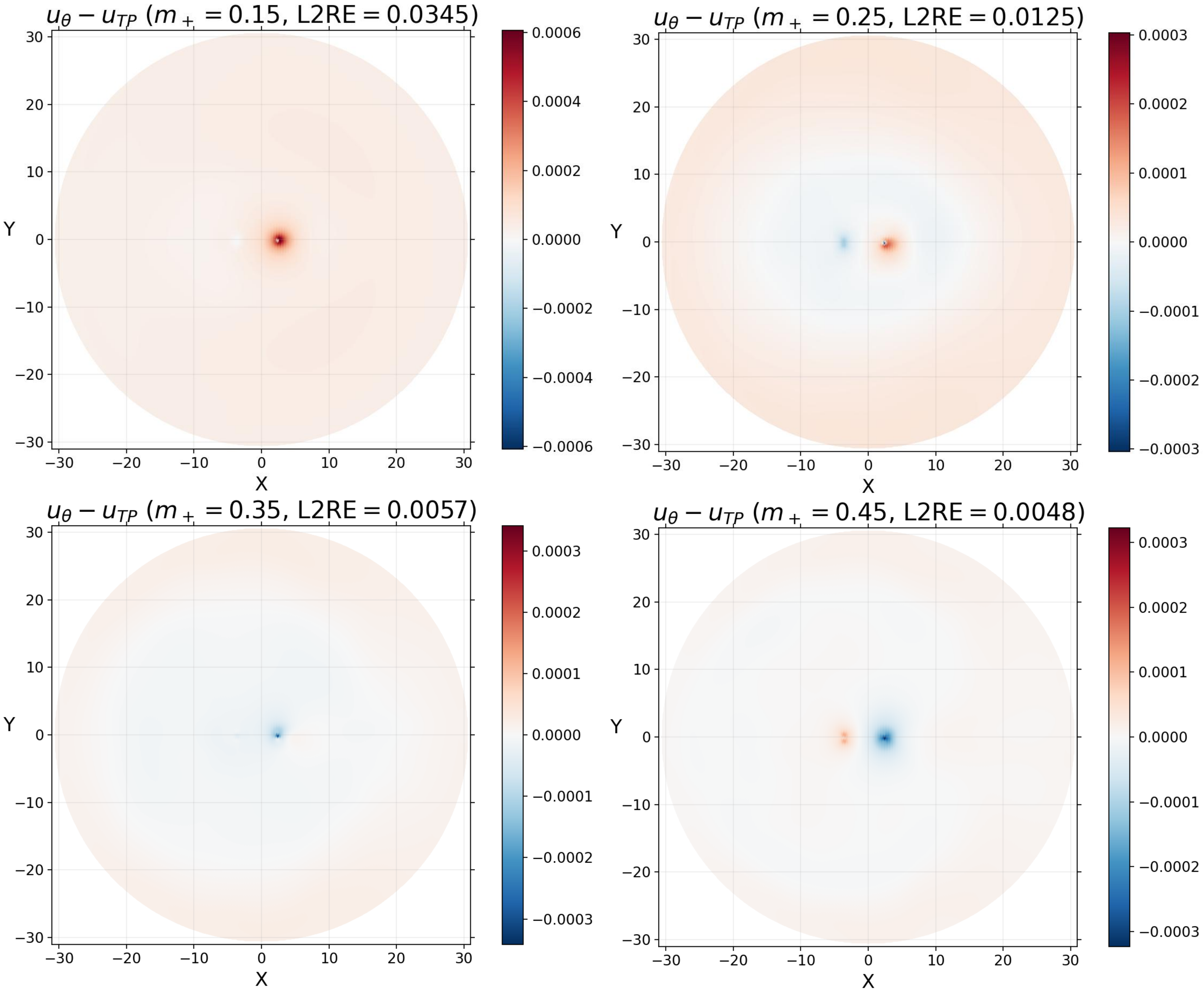
| | L2RE |
|---|---|
| Validation dataset | 0.0202 |
| Test dataset | 0.0190 |
| $m_+ = 0.15$ | 0.0501 |
| $m_+ = 0.25$ | 0.0204 |
| $m_+ = 0.35$ | 0.0085 |
| $m_+ = 0.45$ | 0.0046 |

# Loss graphs

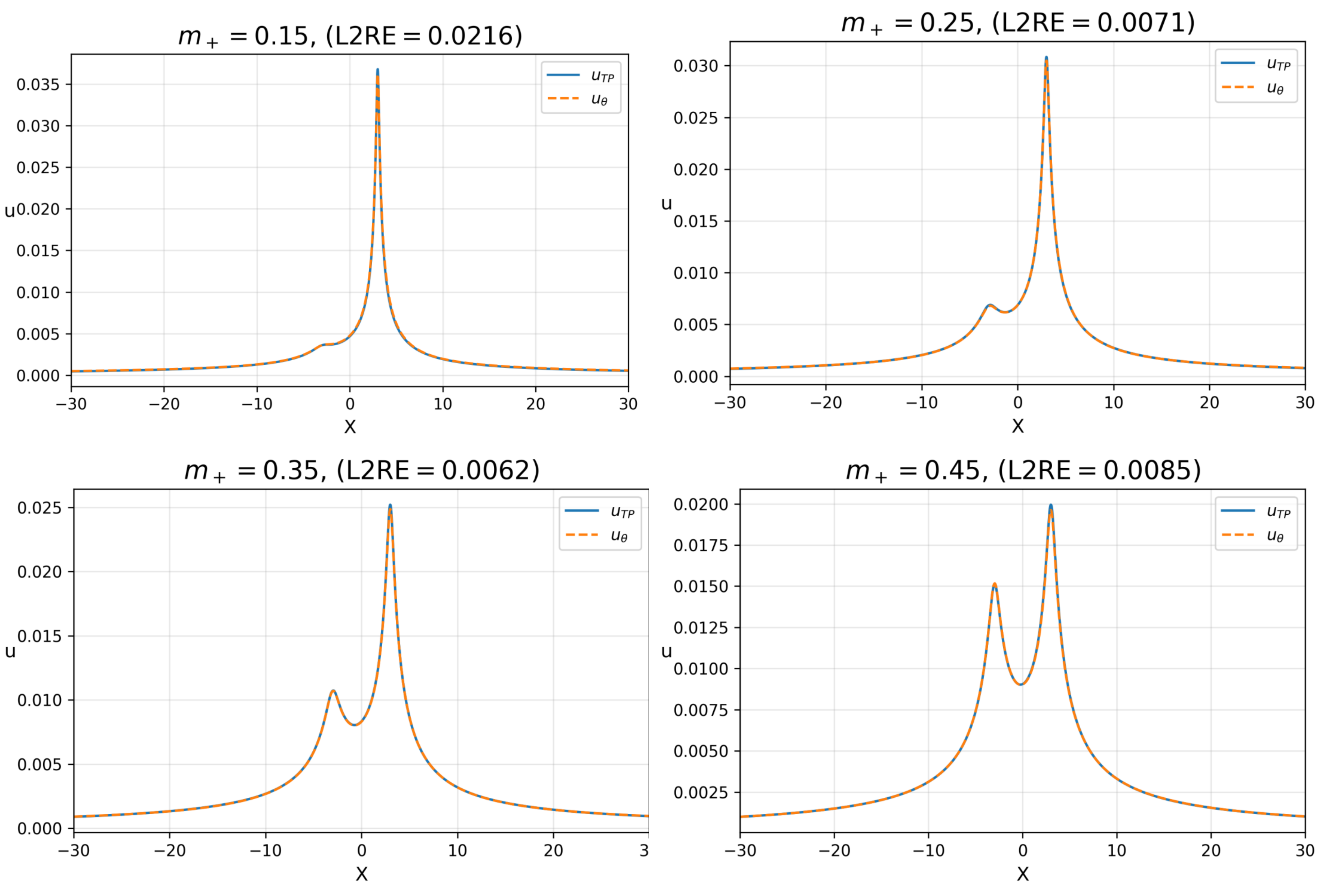


Loss during training
Loss
$L_{data}$
$L_2$
soft-$L_{inf}$
$L_{BC}$
$10^0$
$10^{-2}$
$10^{-4}$
$10^{-6}$
$10^{-8}$
$10^{-10}$
$10^{-12}$
Validation loss
Loss
L2RE
0.07
0.06
0.05
0.04
0.03
0.02
0
500
1000
1500
2000
2500
Epoch

$\times$ 2.5 resolution (2D)



$\times$ 25
res
(1D)



# Conclusion

- Low L2RE on validation (0.0202) and testing (0.0190) datasets shows proof of concept
- Lower $m_+$ values lead to larger deviations from paired $u_{TP}$
  - PIDONet struggles to accurately predict sharp spatial changes in gravitational correction due to lower mass blackholes
- PIDONet able to generate 2D dataset to 2.5 times initial resolution and 1D datasets to 25 times initial resolution to 1% average accuracy

## Next steps:

- Increase the density of collocation points near individual punctures instead of $R < 10$
- Explore PIDONet's ability to predict $u_\theta$ for different puncture locations, positions, and momenta
- Extend the branch net to take into account spinning BBH systems

# Thank you for listening